Run report: two orbit columns under the lazy member index

Run 2026-10-18T18:00Z (UTC start of the first session). Observations only; interpretation is in `../../REPORT.pdf`.

Command. `bash research/ecbench_lazy_columns2_20261018/run.sh` (each session: `ecbench run --wait --spec docs/ecbench/specs/claim-koblitz-lazy-c2-a<a>-n<n>.json --cpus none --allow-busy`, then `ecbench verify --replay-all`, `ecbench table`, `ecbench compare --save`). Specs from `make_specs.sh`; protocol `PROTOCOL.md`, committed in `760c672a5` before the run; sessions record that commit (`git_dirty false`) and binary `ae108d54…`.

Host and isolation. 4-vCPU Intel Xeon VM, Linux x86-64, `--cpus none`: isolation L0 for every record. Counts are recorded at any level; wall times are not admitted.

Arms (all `*_full`): `rho-frob` (reference), `rho-strong`, `ic-lazy-c<c0>` (`compact-orbit-scan:columns=<c0>`, `mitm-frobenius-lazy-counted:m=2`, `stride-renew`; c0 = 1, 1, 3, 1, 8 for n = 13, 17, 23, 29, 37), `ic-lazy-c2` (the same with `columns=2`), `rho-frob-aa` (A/A). Targets per curve 32 (n37: 16), 3 rounds plus one warm-up, public targets, target seeds 202610182001 to 202610182009.

session dir	session id	spec id	records	status	replays reproduced
a0-n13	ECBS1h5f7f47653898	ECS1hbe4a2df36fc5	640	verified:640	480
a1-n17	ECBS1had6c4dfa75fc	ECS1h322728aff7a6	640	verified:640	480
a0-n23	ECBS1h88a016cfa7c0	ECS1h980f02d2fbe5	640	verified:640	480
a1-n29	ECBS1h486ae2ee8d06	ECS1he1726b2155de	640	verified:640	480
a0-n37	ECBS1hbef0622fd5f3	ECS1hce97051a6418	320	verified:320	240

Completed vs requested. 2 880 executions planned and written, all `verified`; no failed, timed-out or exhausted run. 2 160 replays reproduced, 0 problems (`../../audits/`).

Measured values. Paired ratios `frozen c ÷ c = 2` in `ecbench.gae` with the harness's cluster-bootstrap 95 % interval (`../../sessions/*/comparisons/`): n13 0.637 [0.630, 0.645]; n17 1.176 [1.063, 1.297]; n23 0.778 [0.695, 0.874]; n29 0.644 [0.625, 0.665]; n37 0.246 [0.200, 0.309]. `rho-frob-aa ÷ rho-frob` is 1 [1, 1] on every curve. Raw records `../../sessions/<dir>/records.jsonl`; tables `../../tables/`.

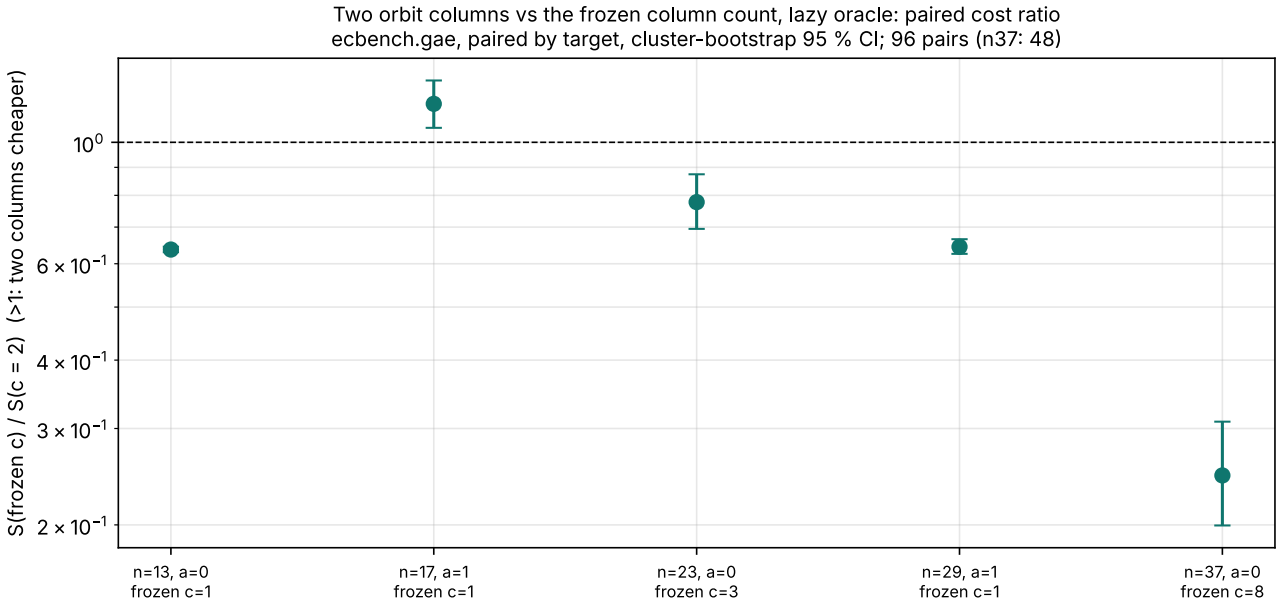

Figure 1: Frozen-c ÷ two-column paired cost ratio per curve, sessions as above.